

X-Code Processing Product Guide

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X-Code Processing Product Guide overview

This guide provides Mastercard customers with an overview of X-Code processing, a Mastercard Network resiliency service that allows certain authorization transactions to be approved when neither issuer processing nor Stand-In processing is available to provide a timely and valid response.

The guide provides an overview of:

- X-Code processing capabilities
- X-Code processing customer onboarding and implementation standards
- Roles and responsibilities of customers and Mastercard

NOTE: Throughout this document, X-Code processing refers to Acquirer MIP X-Code processing.

Audience

The audiences for this document are Mastercard customers, acquirers and issuers, globally and their registered service providers, such as processors, that submit and respond to authorization requests through the Mastercard Network Dual Message System.

Product

X-Code processing is the Mastercard fallback decisioning service for acquirers, which is implemented at an acquirer’s Mastercard Interface Processor (MIP). Refer to the *Product summary* section of this guide for additional details.

Table 1: Details

Audience	Acquirers, Issuers, Processors
Product	X-Code
Region	Global
Country	N/A
Type	Product Guide
Publish Date	15 April 2025

Chapter 1 X-Code processing overview

This chapter contains an overview of X-Code processing, including information about the product offering, the benefits of the product, and how X-Code works.



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Product summary

X-Code processing is the Mastercard fallback decisioning service for acquirers, which is implemented at an acquirer's Mastercard Interface Processor (MIP).

X-Code processing is the tertiary-processing path for online customers, with the primary-processing path being issuer processing and the secondary-processing path being Stand-In processing. For more information about Mastercard Stand-In processing, refer to the *Stand-In Authorization Reference Guide*.

NOTE: Throughout this document, mentions of Stand-In processing refers to either Mastercard Stand-In or an alternative issuer on-behalf authorization service.

X-Code processing is triggered when neither issuer processing nor Stand-In processing is available to provide a timely and valid response to an authorization request. This can happen under two scenarios:

1. The acquirer's MIP is unable to communicate with the Mastercard Network.
2. The acquirer's MIP has sent an authorization request message to the Mastercard Network but neither the issuer nor the Stand-In System provides a timely and valid response to the acquirer's MIP.

Authorization transactions processed through X-Code processing are identified when the Authorization Request Response/0110 message has a value of 000000 in Data Element (DE) 121 (Authorizing Agent ID Code).

Benefits

X-Code processing plays a critical role in the resiliency of the Mastercard Network, providing a third line of defense if both the issuer and the Stand-In System are unavailable for decisioning.

Without X-Code processing, authorization transactions initiated during periods of technical outage may be declined, potentially disrupting the flow of commerce and customer business.

X-Code processing helps ensure business continuity to all parties in the ecosystem:

- Merchants:** For merchants, X-Code processing enables certain purchase transaction completion during outages. This helps merchants provide a smooth check-out experience to cardholders and avoid shopping cart abandonment.
- Acquirers:** For acquirers, X-Code processing may provide the peace of mind that if the acquirer's authorization host can connect to the acquirer's MIP, an acquirer can receive an authorization transaction response from Mastercard.
- Issuers:** By providing a third path of authorization approval, issuers may have peace of mind that their cards may remain top-of-wallet, with reduced disruption to the

cardholder's payment experience. With X-Code processing providing authorization transaction decisioning during outages, issuers benefit from fewer potential operational challenges such as call center inquiries potentially resulting in a better issuer and cardholder experience.

Cardholders: Cardholders may experience fewer disruptions during the checkout process if any outages occur. By providing an authorization response, X-Code processing helps cardholders enjoy a seamless checkout experience with their Mastercard payment method of choice.

How X-Code processing works

This section provides an overview of how X-Code processing works for authorization transactions switched through the Mastercard Network. Note that the Mastercard X-Code processing applies only to Mastercard-branded card programs. X-Code processing declines all authorization transactions from non-Mastercard cards, including private label cards.

Activation of X-Code processing

X-Code processing activates when the acquirer's MIP cannot communicate with the Mastercard Network, or neither the issuer nor the Stand-In System provides a timely and valid response to the acquirer's MIP.

NOTE: As X-Code processing is located at the acquirer's MIP, X-Code processing is only active when the acquirer's host has connectivity with the acquirer's MIP.

X-Code processing rules

The following are rules that are applicable to X-Code processing.

Authorization transaction eligibility

Authorization transactions that are eligible for X-Code processing include card-present and card-not-present purchases from Mastercard debit and credit cards.

A subset of authorization transactions are eligible for X-Code processing and may receive an approval. Authorization transactions ineligible for X-Code processing are always declined.

The following are examples of authorization transaction types that are not eligible for X-Code processing:

- Account Status Inquiry (ASI) requests
- ATM Installment Inquiry requests
- Balance inquiries
- Private label authorization transactions
- Prepaid card authorization transactions
- Product Inquiry Service requests
- ATM and manual cash disbursement authorization transactions

- Payment Transactions, including Gaming Payment Transactions and MoneySend[®]Payment Transactions
- Unique authorization transactions (identified with an authorization transaction category code [TCC] value of U [Unique])

X-Code processing ceiling limits

Ceiling limits are assigned to each product category and authorization transactions with amounts above the ceiling limits are not approved.

The following table lists the maximum approval limit defined by Mastercard for each product category. Within the product category, different approval limits are assigned to each product code.

Table 2: X-Code processing ceiling limits

Product category	Maximum MIP X-Code processing limits
Authorization transactions using Mastercard consumer credit products	USD 950
Authorization transactions using Mastercard consumer debit products	USD 400
Authorization transactions using Mastercard small business commercial products	USD 950
Authorization transactions using Mastercard Corporate commercial products	USD 950
Authorization Transactions using all other Mastercard, Maestro [®] , and Cirrus [®] products	USD 0

X-Code processing risk assessment

For eligible authorization transactions with amounts falling below the ceiling limits, X-Code processing applies risk assessment measures to determine the authorization response, including but not limited to the following:

- Mastercard declines authorization transactions for customers who express a desire to prevent on-behalf approvals by Mastercard.
- Mastercard evaluates each authorization transaction based on merchant types and geography.
- Mastercard monitors the status of account ranges and determines whether each account ranges is eligible for X-Code processing.
- Mastercard monitors authorization transaction velocity and decline authorization transactions if abnormal spending velocity is observed.

These measures are enhanced on an ongoing basis.

X-Code processing of Identity Check authorization transactions

Subject to all previously listed processing requirements, X-Code may approve EMV 3-D Secure transactions. When X-Code approves a Security Level Indicator (SLI) 212 EMV 3-D Secure transaction, X-Code downgrades SLI and responds with the following:

- Data Element (DE) 48, subelement 42, subfield 1 (Electronic Commerce Security Level Indicator and UCAF Collection Indicator) contains the downgraded SLI value.
- DE 48, subelement 42, subfield 2 (Original Electronic Commerce Security Level Indicator and UCAF Collection Indicator) contains the original SLI value.
- DE 48, subelement 42, subfield 3 (Reason for UCAF Collection Indicator Downgrade) contains the new downgrade reason code of value 3 X-Code Processing.

X-Code processing authorization response

When X-Code processing is initiated, the acquirer's MIP generates the following authorization responses:

Table 3: X-Code processing authorization response

MIP X-Code processing authorization response	Type of authorization transaction
91 (Authorization System or issuer system inoperative)	Ineligible authorization transaction types.
92 (Unable to route transaction)	
05 (Do Not Honor)	Authorization transactions that exceed the ceiling limits defined in the MIP X-Code processing Ceiling Limits table. Authorization transactions declined because of MIP X-Code processing risk assessment.
04 (Capture Card)	Authorization transactions to accounts that are listed on the Electronic Warning Bulletin file.
00 (Approve)	Authorization transactions that are less than or equal to the MIP X-Code processing ceiling limits and do not meet any of the criteria previously listed.
87 (Purchase Amount Only, No Cash Back Allowed)	Approval responses provided by X-Code processing to purchase with cash back authorization transactions, if the approval criteria are met.

Liability for X-Code processing authorization transactions

Issuers retain authorization-related liability for X-Code processing-approved authorization transactions up to the applicable X-Code processing ceiling limit, except when X-Code processing approval results from a merchant's resubmission of an issuer-declined card-not-present authorization transaction containing a "do not try again" merchant advice code (a value

of 03 in DE 48, subelement 84). X-Code processing approval has no impact on the fraud-related liability of an authorization transaction. Refer to the *Chargeback Guide* for information about authorization-related and fraud-related chargebacks and other dispute resolution procedures.

Mastercard recommends that issuers contact Mastercard Customer Support or Mastercard Dispute Resolution representatives before raising or initiating any chargebacks related to X-Code processing.

Chapter 2 Use cases and user flows

This chapter describes the transaction flow of X-Code processing under different situations.

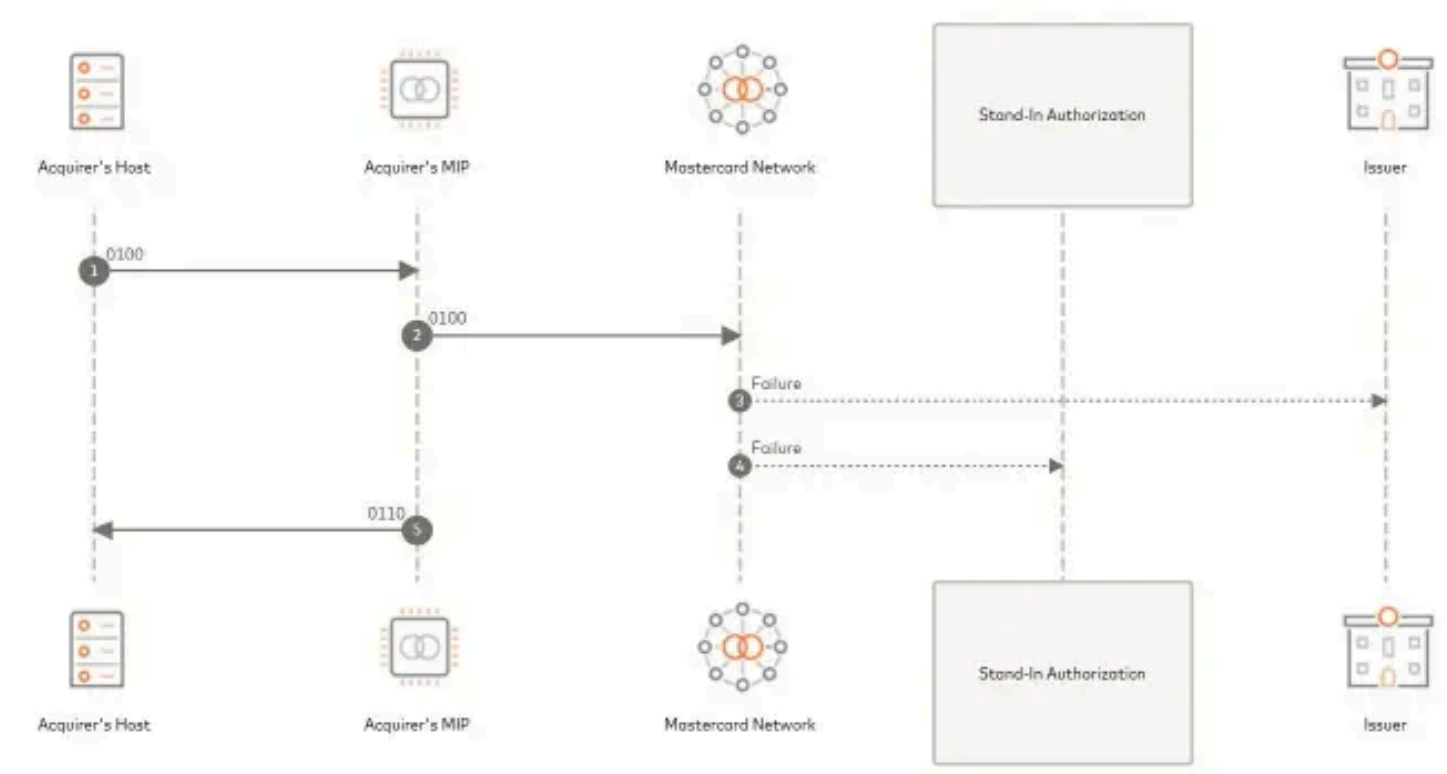
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X-Code processing flow

There are two different scenarios when X-Code processing may be triggered.

Scenario 1: The acquirer's MIP can connect to the Mastercard Network, but neither the issuer nor the Stand-In System provides a timely and valid response to the Mastercard Network.

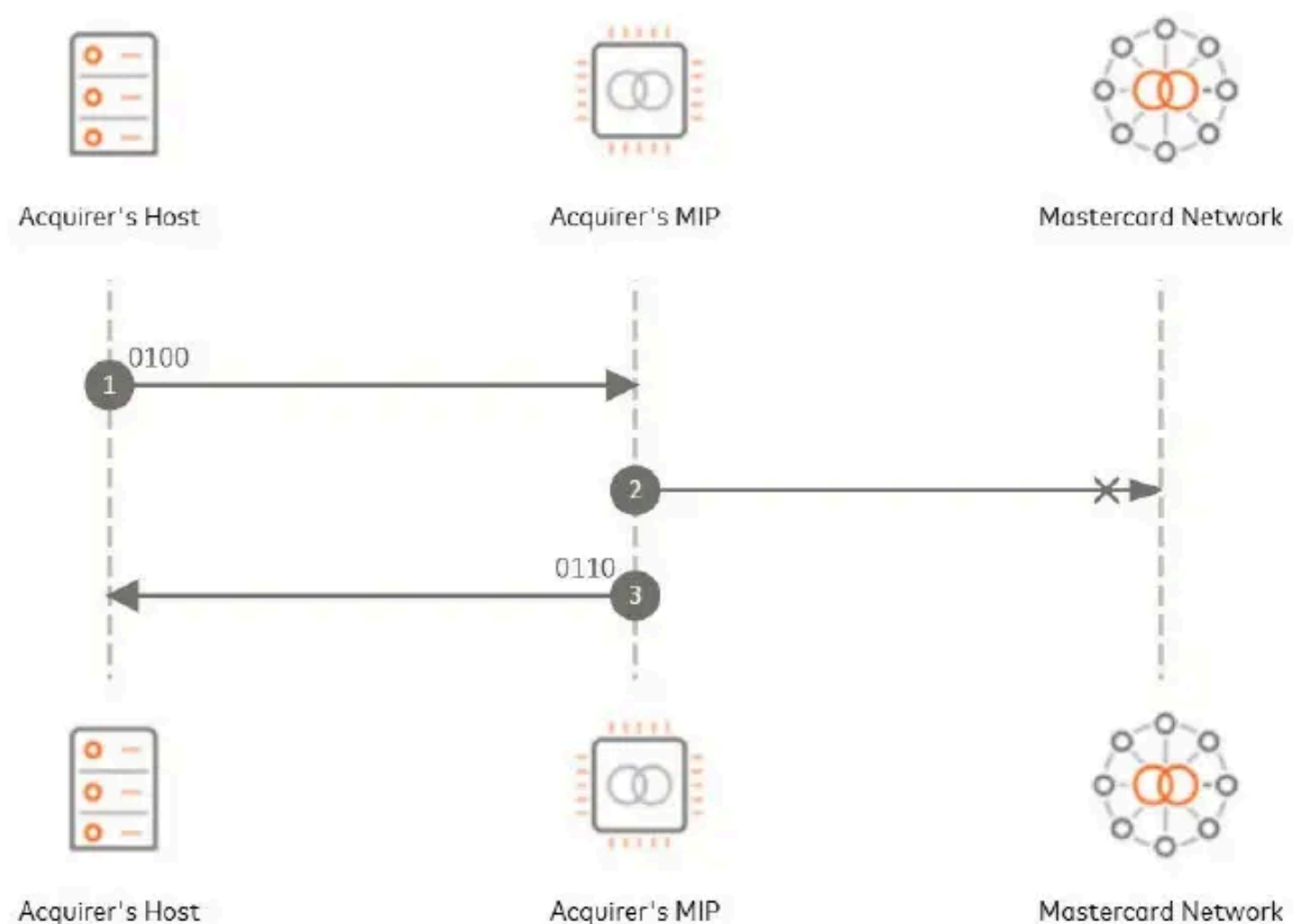
Figure 1: X-Code processing scenario 1 flow



1. The acquirer's host sends an Authorization Request/0100 message to the acquirer's MIP.
2. The acquirer's MIP sends the Authorization Request/0100 message to the Mastercard Network, where certain actions are performed such as fraud score calculations.
3. The Mastercard Network does not receive a valid response from the issuer.
4. The Mastercard Network does not receive a valid response from the Stand-In System.
5. The acquirer's MIP approves or declines the authorization transaction through X-Code processing and sends an Authorization Request Response/0110 message to the acquirer's host.

Scenario 2: Acquirer's MIP cannot communicate with Mastercard Network:

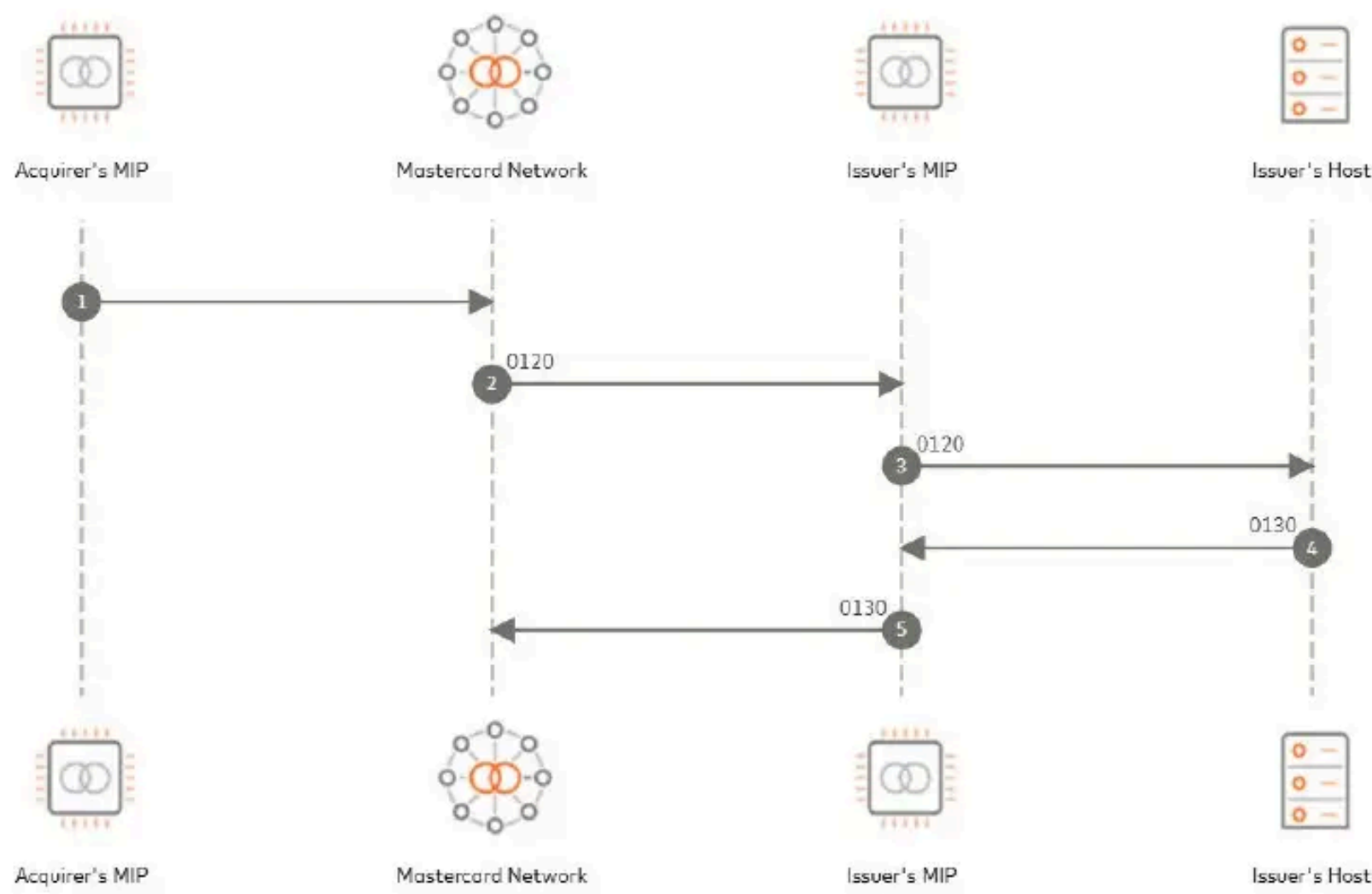
Figure 2: X-Code processing scenario 2 flow



1. The acquirer's host sends an Authorization Request/0100 message to the acquirer's MIP.
2. The acquirer's MIP is disconnected from the Mastercard Network.
3. The acquirer's MIP approves or declines the authorization transaction through X-Code processing and sends an Authorization Request Response/0110 message to the acquirer's host.

Following either of the aforementioned two scenarios, after the connection restores, issuers are notified of the outcome of X-Code processing through an Authorization Advice/0120 message as described in *Outcome of X-Code processing through an Authorization Advice/0120 message flow*.

Figure 3: Outcome of X-Code processing through an Authorization Advice/0120 message flow



1. The acquirer's MIP sends the X-Code processing authorization transaction details to the Mastercard Network.
2. The Mastercard Network generates an Authorization Advice/0120 message for each X-Code processed authorization transaction and delivers the Authorization Advice/0120 message to the issuer's MIP.
3. The issuer's MIP delivers the Authorization Advice/0120 message to the issuer's host.
4. The issuer's host responds with an Authorization Advice Response/0130 message to the issuer's MIP which confirms receipt of the Authorization Advice/0120 message.
5. The issuer's MIP delivers the Authorization Advice Response/0130 message to the Mastercard Network.

Chapter 3 Reporting

This chapter includes information about X-Code processing-related Operational Reports. These reports are available for customer sign-up if customers are interested in receiving more data from Mastercard about X-Code processing.

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Authorization Summary Report (AB505010-AA)

This report shows authorization transaction processing volumes and percentages by BIN for the week to provide customers with a tool to analyze authorization performance.

Specifically, the report includes a MIP Processed Activity Summary of all authorization activity processed as X-Code processing authorization transactions at the acquirer's MIP.

This report is delivered online, in bulk, and in paper formats.

The audience of this report includes acquirers, issuers, and their processors.

For more information about the report layout including a sample report, refer to the *Mastercard Network Processing Dual Message Authorization System Guide*.

For information about the pricing of this report, refer to the Pricing and Billing Resource Center on Mastercard Connect®.

X-Code and Basic Authorization Truncation Weekly Detail Authorizations Report (AB110010-AA)

This report provides details pertaining to authorization transactions authorized by X-Code processing and Basic Authorization Truncation (BAT) systems.

BAT offers more details compared to the Authorization Summary Report including cardholder account number level information.

This report is delivered online, in bulk, and paper formats.

The audience of this report includes issuers and their processors.

For information about the pricing of this report, refer to the Pricing and Billing Resource Center on Mastercard Connect.

Sign up for reports

To sign up for the Operational Reports, contact Customer Support or directly online through Mastercard Connect.

Chapter 4 X-Code processing product onboarding

This chapter provides an overview of the customer onboarding process of X-Code processing for acquirers and issuers.



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X-Code processing onboarding requirements

X-Code is an integral resiliency tool in the Mastercard Network that benefits both acquirers and issuers but involves different implementation for each.

Acquirers

Acquirers automatically receive X-Code when they sign up with Mastercard and are assigned a MIP. No additional enrollment or onboarding is required.

Issuers

Issuers are not required to take any action regarding X-Code. Issuer authorization transactions are automatically subjected to X-Code, a function embedded in an acquirer MIP, as described in this guide.

Mastercard understands there may be certain scenarios where an issuer's authorization transactions should not be subject to X-Code approval. Mastercard may, in its sole discretion, review such scenarios on a case-by-case basis to determine whether an exception to X-Code is valid. To request an exception, contact your Mastercard representative.

Chapter 5 Testing and implementation

This chapter describes the implementation and customer support process of X-Code processing.

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Implementation milestones

X-Code processing implementation is included in the Mastercard standard customer onboarding procedures. No additional steps are required for customers to implement X-Code processing.

The following are the stages and processes associated with a standard implementation plan.

Figure 4: Stages and processes of Mastercard customer onboarding implementation



1. Project initiation
2. Mastercard Test Facility (MTF) onboarding
3. Mastercard Test Facility (MTF) testing
4. Production onboarding
5. Production testing
6. Production live

X-Code processing test cases

X-Code processing test cases are available in the *Issuer Credit and Debit Test Cases* manual located on the Technical Resource Center within Mastercard Connect.

X-Code processing test cases include issuer processing of Mastercard system-generated X-Code processing authorization advices.

X-Code processing customer support

Production support process

The Mastercard Global Customer Service team handles all inquiries and escalations regarding X-Code processing.

1. **Customer contacts Mastercard**

The recommended method of contact is by telephone. Email messaging is also available for noncritical or not urgent issues.

NOTE: For critical issues, report by telephone.

Table 4: Global Customer Service

Method of contact	Contact details
Customer Service Support Line	+1-800-999-0363
Email address (noncritical issues)	Customer_support@mastercard.com

Table 5: Mission Control and Operations (MC&O) Center: critical issues only

Method of contact	Contact details
Phone	+1-636-722-6220 Toll-free: 800-358-3060
Email address	occ@mastercard.com

To ensure the appropriate research and to ensure timely response, provide the required information:

- Customer name
- Date and specific time of occurrence and time zone
- Application impacted: X-Code processing
- Description of issue
- Transaction attributes (BIN last four digits, ARN, Date, Amount, and any other relevant information)

2. **Case created and assigned a priority**

Customer Service creates a case, assigns a tracking number and a priority, and sends a communication back to the customer.

For escalation, call the Customer Service support line or use the **Escalate Now** link in the case creation email notification.

Expected customer communication response time from Mastercard to the customer:

Table 6: Escalation priority

Priority	Initial engagement	Update frequency
Critical	Response provided on inbound phone call	Every 3 hours
Major	1 business day	Every business day
Minor/Low Impact	2 business days	Every 2 business days

3. Mastercard teams engaged for resolution

Customer Service engages all necessary teams at Mastercard to resolve the issue. There is ongoing communication between Mastercard and the customer between steps 2 and 3.

4. Issue resolved

The customer receives notification when the issue is resolved.

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